

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, MOHALI

2nd Mid-SEMESTER 2014-2015

CHM201: SPECTROSCOPIC AND OTHER PHYSICAL METHODS

FULL MARKS 40

DURATION 1 HR

1. Find the ground state term symbols and their corresponding L, S and J values for Cr^{3+} and Zn^{2+} . **3 + 3 = 6**

2. Term symbols for d^5 are following. Arrange according to their energy in an increasing order. **4**

$^2S, ^2P, ^2D(\text{three}), ^2F(\text{two}), ^2G(\text{two}), ^2H, ^2I, ^4P, ^4D, ^4F, ^4G, ^6S$

3. Schematically demonstrate the number of electronic transitions possible from 2S to 2P states in absence and presence of weak magnetic field. **3+3 = 6**

4. Derive the structure of a molecule A_2B_2 for which the Infra-Red and Raman lines are given below. Assign the vibrational modes for each spectral lines. **1+5**

Frequency (cm ⁻¹)		
2976	Raman (str)	
2703		IR(str, PR contour)
1285	Raman (v str)	
526		IR (v str, PQR)
400		IR (w)

5. A number of possible transitions in the beryllium atom are listed below.

(i) Find out the term symbols which are invalid and state the reason in one sentence.

(ii) Which are "fully allowed"? For those which are not allowed, state the appropriate selection rule in one sentence. **3 + 6 = 9**

(a) $2s5s (^1S_0) \rightarrow 2s5d (^1D_2)$

(b) $2s5s (^3S_1) \rightarrow 2s2p (^1P_1)$

(c) $2s5s (^1S_0) \rightarrow ns^2 (^1S_0)$

(d) $2s5p (^3P_1) \rightarrow 2s3s (^3S_1)$

(e) $2s3p (^3P_1) \rightarrow 3p^2 (^3D_2)$

(f) $2s3p (^3P_0) \rightarrow 3p4p (^3D_1)$

6. Experimental spectral data for 1,2-dibromoethene are given below. Predict the correct geometric isomer (*E* or *Z*) of the molecule from the data and state reason in support. **1 + 2 = 3**

IR	Raman
3161 w	3158 w
3064 s	
2826 w	3061 s
2727 vw	
2630vvw	
2166 vw	
2001 vvw	
1585 s	1581 s
1540 m	
1489m	1485 vw
1369 w	
1328 w	
1254 s	
1150 vw	1150 s
1044 vvw	1044 vw
867 vvvv	862 w
830 v	
748 vs	
738 sh. m.	
580 s	580 vs
465 s	462 m
	368 m
	118 vs